Gaurav Gupta

Principal Engineer — Network Automation, AI & Solution Architecture

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SUMMARY

15+ years in wireless telecom, from field drive testing to leading AI-first automation for one of the largest wireless networks in the world. Every initiative self-initiated: gap identified → scoped → built → shipped to production. Sets RAN automation strategy across open-loop, human-assisted closed-loop and closed-loop modes; first on the team to bring the Claude API and Claude Code into engineering work. Telecom-grown, industry-agnostic.

15+	12	75	~1,860
years in wireless telecom	production automation tools	custom AI skills & agents	engineering hours saved / year

EXPERIENCE

Principal Engineer, Systems Engineering — AI-First Tooling

2024 – Present

Verizon

- First on the team to bring the **Claude API and Claude Code** into engineering work: created **75 custom skills and AI agents** and drove team-wide adoption. Introduced mock-first development, cutting tool-build cycles ~83% (14 days → 3).
- Designed an **MCP connector layer** linking LLMs to network data stores, Jira, a 10,000+ page knowledge base and collaboration tools, with a composable bring-your-own-code design for self-serve automation.
- Built KPI anomaly engines: 4G Z-score against a same-weekday 4-week baseline, 5G UL residual BLER tracking, a vendor-transition tracker, and 5G far-anchor ENDC blacklisting with market-scale recommendations.
- Delivered the team's first self-service analytics platform on GCP BigQuery, an AI fixed-wireless (FWA) diagnostic agent, and a data-quality QA suite with a web dashboard.

Principal Engineer, Systems Engineering — Automation at Scale

Aug 2021 – Present

Verizon (converted from contractor; earlier titles System Architect and Solution Architect, RAN system performance)

- Conceived and led the team's first production **Python automation suite (12 tools)**: 4G/5G CM/PM ingestion, KPI anomaly detection, vRAN vDU KPI analysis, gNB alerts, PCAP parsing, BigQuery analytics and multi-vendor monitoring. With the AI skills and agents, saves **~1,860 engineering hours a year** (~0.9 FTE); became the team standard.
- Set the RAN automation strategy across open-loop, human-assisted closed-loop and closed-loop modes; led tool roadmaps from concept to MVP for vRAN and C-Band.
- Led root-cause analysis on critical cross-domain issues with Solution Architects, Product Owners, dev teams and 10+ vendors; executive readouts; represented Verizon Network Systems at the Super Bowl LVIII command center (Feb 2024).
- Mentored engineers across time zones; set Agile cadence, Jira/Confluence standards and code reviews.

System Performance Engineer — Data Engineering

Aug 2019 – Jul 2021

Verizon (Contractor)

- Architected and delivered the first vRAN performance-counter monitoring pipeline for the Samsung 5G network, from specs and vendor negotiation through production (~5,000 daily reports); extended to a 3-vendor 4G/5G design with a new UI.
- Built an event-monitoring system (pipeline and self-serve GUI) for venue-level monitoring, later used at the Super Bowl command center.
- Tested a next-gen UI and database migrations; wrote engineering specs (decoders) for vendor software releases.

Senior NPO Engineer — Performance Engineering → ML **Sep 2017 – Jul 2019**
Nokia

- Led the Service Performance program migrating 5K nodes from Alcatel-Lucent to Nokia AirScale (~2,000 sites in Rocky Mountain), leading 10 system performance engineers.
- Identified and led ML-based layer-management automation at a Tier 1 carrier as domain SME, requirements to production, including a POC with a cross-functional team of 8; introduced a Tableau dashboard for post-rollout gains.

NPO Engineer — RF Optimization, Greenfield LTE **2013 – 2017**
Nokia

- Delivered greenfield LTE builds, carrier adds, VoLTE launches and RF optimization; commissioned multi-vendor sites with test plans and pass/fail criteria.
- Within two years, led 6 engineers and 4 drive teams on large multi-site rollouts.

RF Tools Engineer · Associate RF Engineer — Device & Drive Testing **2009 – 2012**
Accuver Americas · Avion Systems

- Hands-on device testing, drive testing and log analysis in the field: the ground-truth RF understanding behind every system built since.

SKILLS

AI / ML: Claude API, Claude Code (custom skills & agents), Model Context Protocol (MCP), LLM & agentic workflow design, prompt & context engineering, anomaly detection, human-in-the-loop design, scikit-learn

Programming: Python (pandas, NumPy, FastAPI, Flask, Streamlit, Playwright), SQL, JavaScript / Node.js, Bash, VBA

Cloud & Data: GCP BigQuery (WIF/OIDC), S3, REST / SFTP APIs, Tableau, Grafana

Telecom & RAN: Samsung / Ericsson / Nokia / Corning 4G & 5G, 5G NR (NSA/SA), vRAN, C-Band, LTE, VoLTE, Layer 3 call flows, CM/PM analytics, PCAP analysis

Methods: Agile / Scrum, SAFe, Git, Docker, Ansible, Jira / Confluence / Rally, vendor & cross-functional leadership, executive readouts

EDUCATION

M.S., Electrical Engineering
Lamar University, Beaumont, TX · 2007–2008

B.E., Electronics & Communication Eng.
MITM, Indore, India · 2003–2007

CERTIFICATIONS & LEARNING

AI / LLM Integration & Claude Code
· Anthropic · 2024–now

Data Strategy · UC Berkeley Executive Education · 2022

Project Management Professional (PMP) · PMI · 2021

SAFe 5 Practitioner · Scaled Agile · 2020

Full project portfolio: telecom-guy.github.io